Jaliya Nimantha

Electronic and Telecommunication Engineering Graduate

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Summary

Electronic and Telecommunication Engineering graduate of the University of Moratuwa, Sri Lanka with research experience at the Augmented Human Lab, National University of Singapore. Skilled in deep learning, edge AI deployment, and multi-agent systems, with hands-on work in TinyML, Vision-Language Models, and LLM optimization for resource-constrained devices. Currently focused on Brain–Computer Interfaces, multimodal AI, assistive technologies for people with impairments, controllable LLM generation, and Human–Computer Interaction research. Passionate about building interpretable, human-centered AI systems that combine multimodal sensing and intelligence to improve accessibility, independence, and quality of life.

Work Experience

Augmented Human Lab - National University of Singapore On-site
Research Engineer Intern *Dec 2024 – June 2025*

- Developed and deployed machine learning models on edge devices using TinyML and MediaPipe, enabling real-time on-device inference with minimal computational resources.
- Successfully implemented large language models (LLMs) on low-power devices without GPU acceleration using the llama.cpp framework.
- Worked with LangGraph to develop an agentic framework for structured, autonomous interaction in multi-agent systems.

Augmented Human Lab - National University of Singapore Remote
Research Attachment *Aug 2024 – Dec 2024*

- Worked on fine-tuning Vision-Language Models (VLMs) for textbook-based contextual understanding, enhancing educational accessibility for visually impaired individuals in the Singaporean education system.
- Focused on optimizing multi-modal learning pipelines and developing an agentic framework for autonomous interaction.

Education

University of Moratuwa Moratuwa, Sri Lanka
B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering — Second Class Upper Division; Minor in Mathematics *2022 – 2026*

- **CGPA:** 3.66/4.00
- **Dean's List:** Semester 1, Semester 6, Semester 7, Semester 8
- **Core Modules:** Advances in Computer Vision, Deep Learning for Vision, Neural Networks and Fuzzy Logic, Pattern Recognition, Linear Algebra, Image Processing and Machine Vision, Data Structures and Algorithms, Introduction to Engineering Optimization, Applied Statistics, Digital Signal Processing, Robot Design and Competition, Wireless Communications

Bandaranayake College Gampaha, Sri Lanka
GCE Advanced Level - Physical Science Stream *Jan 2011 – Aug 2019*

- Z-Score: **2.3605** (3 A's) | Island Rank: **287**

Projects

Fusion of Speech and EEG signals for neural speech tracking (Final Year Project) *April 2026*
Investigating neural speech tracking by fusing EEG signals with speech features for improved brain-computer interface applications.

- Signal processing pipelines for EEG-speech feature extraction and alignment.
- Deep learning models to decode speech-related neural activity patterns.

GGUF Converter for Hugging Face* *Dec 2024*
Python CLI tool automating the full pipeline for converting Hugging Face LLaMA models to GGUF format for llama.cpp.

- Quantization with precision options (Q4_0, Q5_1, TQ1_0) and upload to Hugging Face Hub.
- Python CLI and API wrapping llama.cpp's C++ backend for automated build, convert, and server workflows.

(Research collaboration projects with AiSee and Augmented Human Lab)*

AiSee LLaMA Android SDK* *Dec 2024*
Android SDK enabling on-device LLaMA inference with Hugging Face model search and GGUF model loading.

- Load .gguf models from URI with dynamic system prompts and LiveData responses.
- JNI bindings wrapping llama.cpp's C++ inference engine, bridging native performance to Java/Kotlin.

(Research collaboration projects with AiSee and Augmented Human Lab)*

TOPA Extension – Multi-View Temporal Ensembling for Video Understanding *2025*
Extended the TOPA framework (NeurIPS 2024 Spotlight) with multi-view temporal ensembling to improve LLM-based zero-shot video understanding without retraining.

- Multi-view inference sampling K different 10-frame views per video and averaging logits for robust temporal coverage.
- Improved accuracy on MVBench and video captioning benchmarks with no additional training cost.

TransX - Transformer Maintenance Platform *July 2025*
Full-stack platform for electrical transformer maintenance with AI-powered thermal image anomaly detection.

- YOLOv8-based fault detection identifying loose joints, point overloads, and potential faults from thermal images.
- React/TypeScript frontend, Spring Boot backend, Flask ML microservice, and MySQL database integration.

Campus-Yamu *Aug 2024 - Present*
AI-powered platform simplifying university selection for Sri Lankan students by analyzing Z-scores, districts, and preferences using RAG models.

- LangChain and OpenAI integration providing personalized recommendations from UGC Handbook.
- Filters 300+ degree programs in seconds, guiding 200,000+ school leavers annually.

GOYAMA (Rice Disease Prediction) *June 2024*

PyTorch deep learning model achieving 98% accuracy in identifying rice leaf diseases, helping farmers detect crop infections early.

- Detects Bacterial blight, Blast, Brown spot, and Tungro from leaf photos.
- Transfer learning with VGG architecture and Albumentations data augmentation.

Deep Learning with PyTorch: Image Segmentation *May 2024*

U-Net architecture implementation for semantic segmentation, enabling pixel-level labeling for autonomous driving scene understanding.

- TensorFlow/Keras U-Net trained on CARLA self-driving car dataset.
- Pixel-perfect segmentation masks for road, vehicles, and pedestrians.

BioSense-AI *Aug 2023*

A portable health monitoring device with advanced ECG technology for precise diagnostics, connected to a mobile app for real-time health insights.

- Machine learning models to accurately classify conditions like asthma and heart failure.
- Integration of simultaneous ECG and PPG signals for improved health predictions.

Skills

Languages: English (professional), Sinhala (native)
Programming: Python, C++, Java, Matlab, Kotlin
Data Science/AI: Numpy, Pandas, PyTorch, Tensorflow
Software/Design: Altium Designer (PCB), Android Studio, Xcode, Photoshop, Illustrator, Premiere Pro
Frameworks: NodeRED, LangGraph
Other: Version Control (Git), LaTeX

Awards and Scholarships

Semi-Finalists - Meta Llama Incubator Programme* 2025
Meta AI
(Research collaboration project with AiSee and Augmented Human Lab)*

Finalist - Brainstorm 2024 Competition 2024
IEEE EMBS Students Chapter, UOM

Finalist and Best Pitch - ICE 2024 2024
Entrepreneurship Hackathon

3rd Place - MECHA 2023 Competition 2023
Faculty of Medicine, UOC

President Scout Award 2019
Highest rank in Sri Lanka Scout Association

Leadership Experience

Electronic Club, University of Moratuwa 2025 – 2026
Vice President

- Leading student engagement activities, department outreach programs, and student representation in administrative matters.

ICE 2025 - AI Innovation Workshop 2025
Workshop Mentor

- Mentored 30 participants in Sri Lanka's premier AI innovation workshop, guiding teams to build breakthrough prototypes within three days.

Scout Troop, Bandaranayake College 2018
Troop Leader

- Led a troop of over 600 scouts, organizing national-level events including an all-island hiking competition with multiple championship wins.
- Played key role in centenary celebration projects; awarded President Scout Award for outstanding leadership and community service.

References

Prof. Ranga Rodrigo
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